Contents

1 Topological Space

1.1 Basic definition

Definition 1.1.1

Let X be a non-empty set. A set $\mathcal{T}$ of subsets of X is said to be a ***topology*** on X if

(i) X and the empty set, $\varnothing$, belong to $\mathcal{T}$,
(ii) the union of any (finite or infinite) number of sets in $\mathcal{T}$ belongs to $\mathcal{T}$, and
(iii) the intersection of any two sets in $\mathcal{T}$ belongs to $\mathcal{T}$.

The pair $(X, \mathcal{T})$ is called a ***topological space***.

Definition 1.1.2

Let $(X, \mathcal{T})$ be any topological space. Then the members of $\mathcal{T}$ are said to be ***open sets***.

Proposition 1.1.3

If $(X, \mathcal{T})$ is any topological space, then

(i) X and $\varnothing$ are open sets,
(ii) the union of any (finite or infinite) number of open sets is an open set,
(iii) the intersection of any finite number of open sets is an open set.

Definition 1.1.4

Let $(X, \mathcal{T})$ be a topological space. A subset S of X is said to be a ***closed set*** in $(X, \mathcal{T})$ if its complement in X, namely $X \setminus S$, is open in $(X, \mathcal{T})$.

Proposition 1.1.5

If $(X, \mathcal{T})$ is any topological space, then

(i) $\varnothing$ and X are closed sets,
(ii) the intersection of any (finite or infinite) number of closed sets is a

closed set and
(iii) the union of any finite number of closed sets is a closed set.

Definition 1.1.6

A subset $\mathcal{S}$ of a topological space $(X, \mathcal{T})$ is said to be ***clopen*** if it is both open and closed in $(X, \mathcal{T})$.

1.2 Basis for a Topology

Definition 1.2.1

Let $(X, \mathcal{T})$ be a topological space. A collection $\mathcal{B}$ of open subsets of X is said to be a ***basis*** for the topology $\mathcal{T}$ if every open set is a union of members of $\mathcal{B}$.

Remark.
拓扑基不是唯一的，根据定义，一旦一个集族 $\mathcal{B}$ 成为了一个拓扑基，我们无论往 $\mathcal{B}$ 添加多少开集，它还是这个拓扑的基. 因此直觉上，拓扑的基是越小越好用的.

Note.
也就是说，拓扑基是通过任意并运算生成拓扑的对象.

Proposition 1.2.2

Let X be a non-empty set and let $\mathcal{B}$ be a collection of subsets of X. Then $\mathcal{B}$ is a basis for a topology on X if and only if $\mathcal{B}$ has the following properties:
(a) $X = \bigcup_{B \in \mathcal{B}} B$, and
(b) for any $B_1, B_2 \in \mathcal{B}$, the set $B_1 \cap B_2$ is a union of members of $\mathcal{B}$.

Remark.
这里不谈与 X 上的任何一个具体的拓扑的关系，只谈一个集族什么时候能够生成一个拓扑.

Note.
这个命题刻画了一个集族距离成为一组基有多远．即一个集族需要什么额外条件，才能通过任意并这个运算生成出一个拓扑．命题中的条件 (a) 无非对应于 Definition 1.1.1 (i)．而通过基任意并生成的这个操作，天然就是为了 Definition 1.1.1 (ii) 而建立的．因此 (b) 就是为了满足1.1.1 (iii) 而提出的条件．我们发现生成拓扑对于有限交封闭的要求，完全地变成了对于基的有限交的封闭的要求．

Proof. If $\mathcal{B}$ is a basis for a topology on X, say $\mathcal{T}_{\mathcal{B}}$. Then $X \in \mathcal{T}$ means that there is a collection of sets in $\mathcal{B}$,say $\mathcal{B}_1$, such that

$$X = \bigcup_{B\in\mathcal{B}_i} B \subseteq \bigcup_{B\in\mathcal{B}} B$$

And since every sets in $\mathcal{B}$ is a subset of X, it follows that (a) holds. To show (b), we only need to see that $\mathcal{B} \subseteq \mathcal{T}_{\mathcal{B}}$. Since $\mathcal{T}$ is closed under finite intersection, we have $B_1 \cap B_2 \in \mathcal{T}_{\mathcal{B}}$.

Then conversely , if $\mathcal{B}$ has the properties (a), (b), we only show that the collection $\langle\mathcal{B}\rangle$ of union of the members in $\mathcal{B}$, satisfies Definition 1.1.1 (iii). We take $A_1, A_2 \in \langle\mathcal{B}\rangle$, and suppose that

$$A_i = \bigcup_{j\in I_i} B_j, \quad B_j \in \mathcal{B}, \quad I_i \text{ is a index set }, i = 1, 2$$

Then

$$\begin{aligned} A_1 \cap A_2 &= \left(\bigcup_{j_1\in I_1} B_{j_1}\right) \cap \left(\bigcup_{j_2\in I_2} B_{j_2}\right) \\ &= \bigcup_{j_2\in I_2}\left(\left(\bigcup_{j_1\in I_1} B_{j_1}\right) \cap B_{j_2}\right) \\ &= \bigcup_{j_2\in I_2}\left(\bigcup_{j_1\in I_1} (B_{j_1} \cap B_{j_2})\right) \\ &= \bigcup_{j_1\in I_1, j_2\in I_2} B_{j_1} \cap B_{j_2} \end{aligned}$$

Since every $B_{j_1} \cap B_{j_2}$ is a union of members of $\mathcal{B}$, and $A_1 \cap A_2$ is a union of such $B_{j_1} \cap B_{j_2}$, then $A_1 \cap A_2$ is a union of members of $\mathcal{B}$. $\square$

Proposition 1.2.3

Let $(X, \mathcal{T})$ be a topological space. A family $\mathcal{B}$ of open subsets of X is a basis for $\mathcal{T}$ if and only if for any point x belonging to any open set U, there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Remark.
表述 "there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$" 就是在说 "U 可以由 $\mathcal{B}$ 通过 '并' 生成" 的微观讲法. 带入这句话之下, 这个命题几乎就是一句 "废话". 同样的道理也出现在 Proposition 1.2.4

Note.
想要看 $\mathcal{B}$ 是否作为基生成了拓扑 $\mathcal{T}$. 我们不用谈 $\mathcal{B}$ 中的元素如何如何 "并" 出每一个开集 U. 而是只需要 $\mathcal{B}$ 对于开集是足够 "精细" 的, 以至于对于任意的开集 U, 我们可以用 $\mathcal{B}$ 里面的元素在 U 上 "框出" 它的任意一个点.

Proof. If $\mathcal{B}$ is a basis for $\mathcal{T}$. Then we can write U as

$$U = \bigcup_{i \in I} B_i$$

Then for any $x \in U$, there exists $i_0 \in I$ such that $x \in B_{i_0} \subseteq U$.

Now we show the converse. First we show that $\mathcal{B}$ is a basis . We use Proposition 1.2.2 to verify this. Since X is a open set , for any $x \in X$, there is a $B_x \in \mathcal{B}$, such that $x \in B_x \subseteq X$, we get

$$X = \bigcup_{x \in X} B_x \subseteq \bigcup_{B \in \mathcal{B}} B \subseteq X$$

Thus

$$X = \bigcup_{B \in \mathcal{B}} B$$

Take any $B_1, B_2 \in \mathcal{B}$, we know $B_1 \cap B_2$ is again an open set. Then similarly, we get

$$B_1 \cap B_2 = \bigcup_{x \in B_1 \cap B_2} B_x$$

is a union of members in $\mathcal{B}$. Thus $\mathcal{B}$ is actually a basis. Finally , we only need to show that $\mathcal{B}$ generates $\mathcal{T}$, that is $\langle \mathcal{B} \rangle = \mathcal{T}$. $\langle \mathcal{B} \rangle$ is the collection of unions of members in $\mathcal{B}$. $\langle\mathcal{B}\rangle \subseteq \mathcal{T}$ is obvious. $\mathcal{T} \subseteq \langle\mathcal{B}\rangle$ concludes from

$$U = \bigcup_{x \in U} B_x \in \langle\mathcal{B}\rangle$$

□

Proposition 1.2.4

Let $\mathcal{B}$ be a basis for a topology $\mathcal{T}$ on a set X. Then a subset U of X is open if and only if for each $x \in U$ there exists a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Note.
把基 $\mathcal{B}$ 看成是 $\mathcal{T}$ 中代表着 "足够精细" 的某一批. 验证 U 是否是开集, 只需要看 U 是否可以被这一小批集合 "概括".

Proof. $\Longrightarrow$ is immediately from Proposition 1.2.3. Now suppose that for each $x \in U$, ther exists a $B_x \in \mathcal{B}$ such that $x \in B \subseteq U$, then

$$U = \bigcup_{x \in U} B_x \in \langle\mathcal{B}\rangle = \mathcal{T}$$

□

Proposition 1.2.5

Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bases for topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively, on a non-empty set X. Then $\mathcal{T}_1 = \mathcal{T}_2$ if and only if

1. for each $B \in \mathcal{B}_1$ and each $x \in B$, there exists a $B' \in \mathcal{B}_2$ such that $x \in B' \subseteq B$, and
2. for each $B \in \mathcal{B}_2$ and each $x \in B$, there exists a $B' \in \mathcal{B}_1$ such that $x \in B' \subseteq B$.

Proof sketch.
考虑这句话: 表述 "there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$ " 就是在说 "U 可以由 $\mathcal{B}$ 通过 '并' 生成". 1. 是说 $\mathcal{B}_1$ 中任意集合可以由 $\mathcal{B}_2$ "并" 出来. 自然地 $\mathcal{T}_1$ 中任意集合也被 $\mathcal{B}_2$ "并出来". 因此 1. 就是 $\mathcal{T}_1 \subseteq \mathcal{T}_2$. 同样地, 2. 就是 $\mathcal{T}_2 \subseteq \mathcal{T}_1$.

Lemma 1.2.6

Let $\mathcal{B}$ and $\mathcal{B}'$ be bases for the topologies $\mathcal{T}$ and $\mathcal{T}'$, respectively, on X. Then the following are equivalent:

1. $\mathcal{T}'$ is finer than $\mathcal{T}$.
2. For each $x \in X$ and each basis element $B \in \mathcal{B}$ containing x, there is a basis element $B' \in \mathcal{B}'$ such that $x \in B' \subset B$.

Proof sketch.
同样的, 2. 就是在说每个 B 可以由一堆 B' 并出来. 那么显然每个 $U \in \mathcal{T}$ 可以由一堆 B' 并出来, 即 $U' \in \mathcal{T}$.

1.3 Closed Sets and Limit Points

Definition 1.3.1

A subset A of a topological space X is said to be ***closed*** if the set $X - A$ is open.

1.3.1 Closure, Interior

Definition 1.3.2

Given a subset A of a topological space X

- The ***interior*** of A , denoted by A° or by $\operatorname{Int} A$, is defined as the union of all open sets contained in A.
- The ***closure*** of A , denoted by $\bar{A}$ or by $\operatorname{Cl} A$, is defined as the intersection of all closed sets containing A.

Theorem 1.3.3

Let A be a subset of the topological space X.

(a) Then $x \in \bar{A}$ if and only if every open set U containing x intersects A.

(b) Supposing the topology of X is given by a basis, then $x \in \bar{A}$ if and only if every basis element B containing x intersects A.

Proof. (a) Method 1. Note that

$$V \text{ is a closed set containing } A \iff V^c \text{ is an open set such that } V^c \cap A = \varnothing$$

We have

$$\begin{aligned}
x \notin \bar{A} &\iff x \notin \bigcap_{V \text{ is a closed set containing A}} V \\
&\iff x \in \bigcup_{V \text{ is a closed set containing } A} V^c \\
&\iff x \in \bigcup_{U \text{ is an open set such that } U \cap A = \varnothing} U \\
&\iff \text{there exists open set } U \text{ such that } x \in U, U \cap A = \varnothing
\end{aligned}$$

Method 2. Suppose that $x \in \bar{A}$, then x is contained in any closed set containing A . If there exists open set U containing x dose not intersects A. Then U^c is a closed set containing A, which implies that $x \in U^c$, a contradiction. Conversely , if every open set containing x intersects A, then every closed set that x dose not lie in cannot contain A. i.e. x lies in every closed set containing A.

(b) If every open set containing x intersects A, then every basis B containing x intersects A since B is also an open set . Conversely, if every basis B containing x intersects A, since every open set U contains a basis contaning x , it intersects with A as well. Then it follows from (a).

□

1.3.2 Limit Points

Definition 1.3.4

If A is a subset of the topological space X and if x is a point of X, we say that x is a ***limit point*** (or "cluster point," or "point of accumulation") of A if every neighborhood of x intersects A in some point ***other than x itself***. Said differently, x is a limit point of A if it belongs to the closure of $A - \{x\}$.

Theorem 1.3.5

Let A be a subset of the topological space X; let A' be the set of all ***limit points*** of A. Then

$$\bar{A} = A \cup A'.$$

Proof sketch.
对于 $\bar{A} \subseteq A \cup A'$, $A \subseteq \bar{A}$ 显然而 $A' \subseteq A$ 是 Theorem 1.3.3的直接推论. $A \cup A' \subseteq \bar{A}$, 则考虑若 x 不在 A 里面, 那么 $x \in A'$ 的定义中的开集一定与 A 相交, 从而有 $x \in \bar{A}$.

1.4 The Subspace Topology

Definition 1.4.1

Let X be a topological space with topology $\mathcal{T}$. If Y is a subset of X, the collection

$$\mathcal{T}_Y = \{Y \cap U : U \in \mathcal{T}\}$$

is a topology on Y , called the ***subspace topology***. With this topology, Y is called ***subspace*** of X; its open sets consist of all intersections of open sets of X with Y.

Lemma 1.4.2

If $\mathcal{B}$ is a basis for the topology of X then the collection

$$\mathcal{B}_Y = \{B \cap Y \mid B \in \mathcal{B}\}$$

is a basis for the subspace topology on Y.

Lemma 1.4.3

Let Y be a subspace of X. If U is open in Y and Y is open in X, then U is open in X.

1.5 The Product Topology

1.5.1 Product of the Two

Theorem 1.5.1

If $\mathcal{B}$ is a ***basis*** for the topology of X and $\mathcal{C}$ is a ***basis*** for the topology of Y, then the collection

$$\mathcal{D} = \{B \times C \mid B \in \mathcal{B} \text{ and } C \in \mathcal{C}\}$$

is a ***basis*** for the topology of $X \times Y$.

Definition 1.5.2

Let $\pi_1 : X \times Y \to X$ be defined by the equation

$$\pi_1(x, y) = x;$$

let $\pi_2 : X \times Y \to Y$ be defined by the equation

$$\pi_2(x, y) = y.$$

The maps π_1 and π_2 are called the ***projections*** of $X \times Y$ onto its first and second factors, respectively.

Theorem 1.5.3

The collection

$$\mathcal{S} = \{\pi_1^{-1}(U) \mid U \text{ open in } X\} \cup \{\pi_2^{-1}(V) \mid V \text{ open in } Y\}$$

is a ***subbasis*** for the ***product topology*** on $X \times Y$.

1.5.2 The General Product

Definition 1.5.4

Let J be an index set. Given a set X, we define a J-***tuple*** of elements of X to be a function $\mathbf{x} : J \to X$. If α is an element of J, we often denote the value of $\mathbf{x}$ at α by x_α rather than $\mathbf{x}(\alpha)$; we call it the αth ***coordinate*** of $\mathbf{x}$. And we often denote the function $\mathbf{x}$ itself by the symbol

$$(x_\alpha)_{\alpha \in J},$$

which is as close as we can come to a "tuple notation" for an arbitrary index set J. We denote the set of all J-tuples of elements of X by X^J.

Definition 1.5.5

Let $\{A_\alpha\}_{\alpha \in J}$ be an indexed family of sets; let $X = \bigcup_{\alpha \in J} A_\alpha$. The ***carte-***

sian product of this indexed family, denoted by

$$\prod_{\alpha \in J} A_\alpha,$$

is defined to be the set of all J-tuples $(x_\alpha)_{\alpha \in J}$ of elements of X such that $x_\alpha \in A_\alpha$ for each $\alpha \in J$. That is, it is the set of all functions

$$\mathbf{x} : J \to \bigcup_{\alpha \in J} A_\alpha$$

such that $\mathbf{x}(\alpha) \in A_\alpha$ for each $\alpha \in J$.

Definition 1.5.6

Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of topological spaces. Let us take as a basis for a topology on the product space

$$\prod_{\alpha \in J} X_\alpha$$

the collection of all sets of the form

$$\prod_{\alpha \in J} U_\alpha,$$

where U_α is open in X_α, for each $\alpha \in J$. The topology generated by this basis is called the ***box topology***.

Definition 1.5.7

Let S_β denote the collection

$$S_\beta = \{\pi_\beta^{-1}(U_\beta) \mid U_\beta \text{ open in } X_\beta\},$$

and let $\mathcal{S}$ denote the union of these collections,

$$\mathcal{S} = \bigcup_{\beta \in J} S_\beta.$$

The topology generated by the subbasis $\mathcal{S}$ is called the ***product topology***. In this topology $\prod_{\alpha \in J} X_\alpha$ is called a ***product space***.

Theorem 1.5.8 ▶ Comparison of the *box and product topologies*

The box topology on $\prod X_\alpha$ has as basis all sets of the form $\prod U_\alpha$, where U_α is open in X_α for each α. The product topology on $\prod X_\alpha$ has as basis all sets of the form $\prod U_\alpha$, where U_α is open in X_α for each α and U_α equals X_α except for finitely many values of α.

Theorem 1.5.9

Suppose the topology on each space X_α is given by a basis $\mathcal{B}_\alpha$. The collection of all sets of the form

$$\prod_{\alpha\in J} B_\alpha,$$

where $B_\alpha \in \mathcal{B}_\alpha$ for each α, will serve as a basis for the box topology on $\prod_{\alpha\in J} X_\alpha$. The collection of all sets of the same form, where $B_\alpha \in \mathcal{B}_\alpha$ for finitely many indices α and $B_\alpha = X_\alpha$ for all the remaining indices, will serve as a basis for the product topology $\prod_{\alpha\in J} X_\alpha$.

Theorem 1.5.10

Let A_α be a subspace of X_α, for each $\alpha \in J$. Then $\prod A_\alpha$ is a subspace of $\prod X_\alpha$ if both products are given the box topology, or if both products are given the product topology.

Theorem 1.5.11

If each space X_α is ***Hausdorff space***, then $\prod X_\alpha$ is a ***Hausdorff space*** in both the ***box*** and ***product topologies***.

Theorem 1.5.12

Let $\{X_\alpha\}$ be an indexed family of spaces; let $A_\alpha \subset X_\alpha$ for each α. If $\prod X_\alpha$ is given either the product or the box topology, then

$$\prod \bar{A}_\alpha = \overline{\prod A_\alpha}.$$

Theorem 1.5.13

Let $f : A \to \prod_{\alpha \in J} X_\alpha$ be given by the equation

$$f(a) = (f_\alpha(a))_{\alpha \in J},$$

where $f_\alpha : A \to X_\alpha$ for each α. Let $\prod X_\alpha$ have the product topology. Then the function f is continuous if and only if each function f_α is continuous.

2

Continuous Functions

2.1 Continuity of a Function

Definition 2.1.1

Let X and Y be topological spaces. A function $f : X \to Y$ is said to be ***continuous*** if for each open subset V of Y, the set $f^{-1}(V)$ is an open subset of X.

Theorem 2.1.2

Let X and Y be topological spaces; let $f : X \to Y$. Then the following are equivalent:

1. f is continuous.
2. For every subset A of X, one has $f(\bar{A}) \subset \overline{f(A)}$.
3. For every closed set B of Y, the set $f^{-1}(B)$ is closed in X.
4. For each $x \in X$ and each neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subset V$.

Remark.
If the condition in (4) holds for the point x of X, we say that f is ***continuous at the point x***.

Proof. 1. $\implies$ 2. : If f is continuous, take any $x \in f\left(\bar{A}\right)$. Let V be any neighborhood of $f(x)$. Then $f^{-1}(V)$ is a neighborhood of x. We have $f^{-1}(V) \cap A \neq \varnothing$. Then

$$\varnothing \neq f\left(f^{-1}(V) \cap A\right) \subseteq V \cap f(A)$$

Since V is arbitrary, $f(x) \in \overline{f(A)}$. □

2.2 Homeomorphisms

Definition 2.2.1

Let X and Y be topological spaces; let $f : X \to Y$ be a bijection. If both the function f and the inverse function

$$f^{-1} : Y \to X$$

are continuous, then f is called a ***homeomorphism***.

2.3 Constructing Continuous Functions

Theorem 2.3.1 ▸ Rules for constructing continuous functions

Let X, Y, and Z be topological spaces.

(a) (Constant function) If $f : X \to Y$ maps all of X into the single point y_0 of Y, then f is continuous.

(b) (Inclusion) If A is a subspace of X, the inclusion function $j : A \to X$ is continuous.

(c) (Composites) If $f : X \to Y$ and $g : Y \to Z$ are continuous, then the map $g \circ f : X \to Z$ is continuous.

(d) (Restricting the domain) If $f : X \to Y$ is continuous, and if A is a subspace of X, then the restricted function $f|A : A \to Y$ is continuous.

(e) (Restricting or expanding the range) Let $f : X \to Y$ be continuous. If Z is a subspace of Y containing the image set $f(X)$, then the function $g : X \to Z$ obtained by restricting the range of f is continuous. If Z is a space having Y as a subspace, then the function $h : X \to Z$ obtained by expanding the range of f is continuous.

(f) (Local formulation of continuity) The map $f : X \to Y$ is continuous if X can be written as the union of open sets U_α such that $f|U_\alpha$ is continuous for each α.

3 Homeomorphisms

3.1 Subcspaces

Definition 3.1.1

Let Y be a non-empty subset of a topological space $(X, \mathcal{T})$. The collection $\mathcal{T}_Y = \{O \cap Y : O \in \mathcal{T}\}$ of subsets of Y is a topology on Y called the ***subspace topology*** (or the ***relative topology*** or the ***induced topology*** or the ***topology induced on*** Y ***by*** $\mathcal{T}$). The topological space $(Y, \mathcal{T}_Y)$ is said to be a ***subspace*** of $(X, \mathcal{T})$.

4 Limit Points

4.1 Limit Points and Closure

Definition 4.1.1

Let A be a subset of a topological space $(X, \mathcal{T})$. A point $x \in X$ is said to be a ***limit point*** (or ***accumulation point*** or ***cluster point***) of $\boldsymbol{A}$ if every open set, U, containing x contains a point of A different from x.

Note.
A 的极限点 x 是不能与 $A \setminus \{x\}$ "分离开" 的点. 也就是说在每一个开集里面, x 和 $A \setminus \{x\}$ 要么都有在里面的点, 要么都没有.

Proposition 4.1.2

Let A be a subset of a topological space $(X, \mathcal{T})$. Then A is closed in $(X, \mathcal{T})$ if and only if A contains all of its limit points.

Proof sketch.
如果集合 A 是闭的, 那么 A 外面的点 x 都被落在 A 外面的一个开集 U 给包住了, 那么就导致了 x 不是极限点.
如果 A 包含了所有极限点, 那么 A 外面的点 x 都不是极限点, 从而 x 一定能被与 A 无交的一个开集给包住.

Proof. If A contains all of its limit points. Take any $x \in A^c$, then x is not a limit point of A. There exists open set U , containing x contains no point of A. Then $U \subseteq A^c$, which implies that A^c is an open set. Thus A is a closed set.

Now, let A be a closed set. Let x be any limit point of A. If $x \notin A$, then there is a open set $U \subseteq A^c$ such that $x \in U$. By the definition of the limit point , there is a point y of A different from x belongs to U, which is a contradiction to $U \cap A = \emptyset$. □

Proposition 4.1.3

Let A be a subset of a topological space $(X, \mathcal{T})$ and A' the set of all limit points of A. Then $A \cup A'$ is a closed set.

Remark.
The proof shows that : If U is an open set, and A an arbitrary set. Then $U \cap A = \varnothing$ implies that $U \cap A' = \varnothing$.

Proof. Let $p \in X \setminus (A \cup A')$, then p neither a point in A nor a limit point of A. Thus there exists a open set U, such that $p \in U$, $U \cap A = \varnothing$. We now claim that $U \cap A' = \varnothing$, otherwise, we take $q \in U \cap A'$. Then U contains a point of A different from q, which is a contradiction to $U \cap A = \varnothing$. Thus we have

$$U \cap (A \cup A') = \varnothing \implies p \in U \subseteq X \setminus (A \cup A')$$

Then $X \setminus (A \cup A')$ is an open set. □

Definition 4.1.4

Let A be a subset of a topological space $(X, \mathcal{T})$. Then the set $A \cup A'$ consisting of A and all its limit points is called the ***closure of A*** and is denoted by $\overline{A}$.

Definition 4.1.5

Let A be a subset of a topological space $(X, \mathcal{T})$. Then A is said to be ***dense*** in X or ***everywhere dense*** in X if $\overline{A} = X$.

Proposition 4.1.6

Let A be a subset of a topological space $(X, \mathcal{T})$. Then A is dense in X if and only if every non-empty open subset of X intersects A non-trivially (that is, if $U \in \mathcal{T}$ and $U \neq \varnothing$ then $A \cap U \neq \varnothing$.)

Proof. If A is dense in X, then $\overline{A} = X$. Take any non-empty open subset U of X. If $A \cap U = \varnothing$, then $A' \cap U = \varnothing$ from Propoistion 4.1.3 Remark . Thus $U \cap \overline{A} = \varnothing$, then $U \cap X = \varnothing$, which is a contradiction. Thus $A \cap U \neq \varnothing$.

Conversely, if for any $U \in \mathcal{T}$, and $U \neq \varnothing$, we have $A \cap U \neq \varnothing$. Take any $x \in X$. If $x \notin A$, then for any open set $V \in \mathcal{T}$, containing x, we have

$A \cap V \neq \varnothing$. And we have $V \cap (A \setminus \{x\}) \neq \varnothing$. Thus x is a limit point of A. It follows that $x \in A \cup A' = \overline{A}$, $X \subseteq \bar{A}$. Then $X = \overline{A}$. $\square$

5 Connectedness

5.1 Connectedness

Definition 5.1.1

A topological space X is ***connected*** if and only if one of the following equivalent conditions holds:

(i) X cannot be expressed as the union of two disjoint nonempty open sets.

(ii) Every continuous function $f : X \to \{0, 1\}$ is constant, where $\{0, 1\}$ is equipped with the discrete topology.

Proof. If (i) holds then since

$$X = f^{-1}(0) \cup f^{-1}(1)$$

one of the component must be empty . Thus f is constant. Then (ii) holds. If (ii) holds, then for each U, V such that $X = U \cap V$, $U \cap V = \varnothing$, define

$$f := \begin{cases} 1, & x \in U \\ 0, & x \in V \end{cases}$$

then one of the U and V must be empty. □

Definition 5.1.2

A topological space X is said to be ***path connected*** if and only if for all $x, y \in X$ there is a ***path*** that connects x and y.

Theorem 5.1.3

If X is (path) connected and $f : X \to Y$, then fX is (path) connected.

Corollary 5.1.4

Connected and ***path connected*** are topological properties.

Corollary 5.1.5

The quotient of a (path) connected space is (path) connected.

Proof. Since quotient maps are continuous surjections, we know quotients preserve (path) connectedness. □

Theorem 5.1.6

Let X be a space and $f : X \to Y$ be a surjective map. If Y is connected in the quotient topology and if each fiber $f^{-1}y$ is connected, then X is connected.

Proof. Let $g : X \to \{0,1\}$. Since the fibers of f are connected, g must be constant on each fiber of f. Therefore g factors through $f : X \to Y$, and there is a map $\bar{g} : Y \to \{0,1\}$ that fits into this diagram.

$$\begin{tikzcd} & X \arrow[d, "f"] \arrow[dl, bend right, "g"'] \\ \{0,1\} & Y \arrow[l, dashed, "\bar{g}"] \end{tikzcd}$$

But Y is connected and so $\bar{g}$ must be constant. Therefore $g = \bar{g}f$ is constant. □

Theorem 5.1.7

Suppose $X = \bigcup_{\alpha\in A} X_\alpha$ and that for each $\alpha \in A$ the space X_α is (path) connected. If there is a point $x \in \bigcap_{\alpha\in A} X_\alpha$ then X is (path) connected.

Proof. 1. Connected: Let $g : X \to \{0,1\}$, since X_α is connected, g is constant on X_α. Specially, $g|_{X_\alpha} \equiv g(x)$ for each $\alpha \in A$. Thus $g \equiv g(x)$ is constant. Thus X is connected.

2. Take any $y_1, y_2 \in X$, suppose that $y_1 \in X_{\alpha_1}, y_2 \in X_{\alpha_2}$. Since $x \in X_{\alpha_1} \cap X_{\alpha_2}$, there exists path γ_1 on X_{α_1} connecting y_1 and x, and path γ_2 on X_{α_2} connecting x and y_2. We define $\gamma : [0,1] \to X$ to be

$$\gamma(t) = \begin{cases} \gamma_1(2t), & t \in \left[0, \frac{1}{2}\right] \\ \gamma_2(2t-1), & t \in \left[\frac{1}{2}, 1\right] \end{cases}$$

Then it follows from the gluing lemma that γ is continuous, connecting y_1 and y_2. Thus X is path-connected.

□

Theorem 5.1.8

If $A \subseteq X$ is connected, then so dose $\bar{A}$.

Proof. Let $f : \bar{A} \to \{0,1\}$ be any continuous map. Since A is connected, f is constant on A. We may assume that $f(A) = \{0\}$. Since f is continuous, $f^{-1}(0)$ is closed in $\bar{A}$ and contains A, which implies that $f^{-1}(0) \supseteq \bar{A}$. Thus $f(\bar{A}) = \{0\}$, f is constant on $\bar{A}$. It follows that $\bar{A}$ is connected. □

5.2 Constructions and Connectedness

Theorem 5.2.1

Let $\{X_\alpha\}_{\alpha \in A}$ be a collection of (path) connected topological spaces. Then $X := \prod_{\alpha \in A} X_\alpha$ is (path) connected.

Proof.

1. Conneted:

2. Path-connected: For each $a, b \in X$, and each $\alpha \in A$. Since X_α is path connected, there exists a path $p_\alpha : [0,1] \to X_\alpha$ connecting a_α and b_α. Then it follows from the universal property that there exists a unique continuous function $p : [0,1] \to X$ such that $\pi_\alpha \circ p = p_\alpha$. Thus $p(0) = (a_\alpha)_{\alpha \in A} = a$, $p(1) = (b_\alpha)_{\alpha \in A} = b$.

□

5.3 Components and Local Connectedness

Definition 5.3.1

Given X, define an equivalence relation on X by setting $x \sim y$ if there is a connected subspace of X containing both x and y. The equivalence classes are called the ***components*** (or the "connected components") of X.

Theorem 5.3.2

The components of X are ***connected disjoint*** subspaces of X whose union is X, such that ***each*** nonempty connected subspace of X intersects only one of them.

Definition 5.3.3

We define another equivalence relation on the space X by defining $x \sim y$ if there is a path in X from x to y. The equivalence classes are called the ***path components*** of X.

Definition 5.3.4

A space X is said to be ***locally connected at x*** if for every neighborhood U of x, there is a connected neighborhood V of x contained in U. If X is locally connected at each of its points, it is said simply to be ***locally connected***. Similarly, a space X is said to be ***locally path connected at x*** if for every neighborhood U of x, there is a path-connected neighborhood V of x contained in U. If X is locally path connected at each of its points, then it is said to be ***locally path connected***.

Theorem 5.3.5

A space X is locally connected ***if and only if*** for every open set U of X, ***each component of*** U is open in X.

Proof. Suppose X is locally connected. Take any open set U of X, and any component C of U. For each $x \in C$, there exists a connected neighborhodd V of x such that $V \subseteq U$. Since $V \cap C \neq \varnothing$, we have $V \subseteq C$. C is open.

If for every open set U of X and each componet C of U, C is open. Then for each $x \in X$, and each neighborhood V of x. The component C of V

containing x is a open conneted neighborhood of x contained in V. Thus X is connected. $\square$

Theorem 5.3.6

A space X is ***locally path connected if and only if*** for every open set U of X, each path component of U is open in X.

Theorem 5.3.7

If X is a topological space, each path component of X lies in a component of X. If X is locally path connected, then the components and the path components of X are the same.

6

Compatctness

6.1 Basic Definition and Properties

Definition 6.1.1

A collection $\mathcal{U}$ of open subsets of a space X is called an ***open cover*** for X if and only if the union of sets in $\mathcal{U}$ contains X. The space X is ***compact*** if and only if every open cover of X has a finite subcover.

Theorem 6.1.2

If X is compact and $f : X \to Y$ is continuous, then fX is compact.

Theorem 6.1.3 ▶ Bolzano-Weierstrass

Every infinite set in a compact space has a limit point.

6.2 FIP , a dual description

There is a dual description.

Definition 6.2.1

Let S be a collection of sets. We say that S has the ***finite intersection property*** if and only if for every finite subcollection $A_1, \dots, A_n \subseteq S$, the intersection $A_1 \cap \cdots \cap A_n \neq \varnothing$. We abbreviate the finite intersection property by FIP.

Theorem 6.2.2

A space X is compact if and only if every collection of closed subsets of X with the FIP has nonempty intersection.

Proof. Let $\{F_\alpha\}_{\alpha\in A}$ be any collection of closed subsets of X with the FIP. If $\bigcap_\alpha F_\alpha = \varnothing$, then $\{F_\alpha^c\}_{\alpha\in A}$ is an open cover of X, which has a subcover $F_1^c, F_2^c, \cdots, F_m^c$,

such that
$$\bigcap_{k=1}^{m} F_m = \left(\bigcup_{k=1}^{m} F_k^c\right)^c = X^c = \varnothing$$
a contradiction.

Conversely, if every closed collection is with FIP. Then, for each open cover $\{U_\alpha\}_{\alpha\in A}$ of X, if $\{U_\alpha\}_{\alpha\in A}$ dose not have a finite subcover of F_α, $\{U_\alpha^c\}_{\alpha\in A}$ is a closed collection of X with the FIP, which imlies that $\bigcap U_\alpha^c \neq \varnothing$, which is a contradiction to $\bigcup U_\alpha = X$. □

Theorem 6.2.3

Closed subsets of compact spaces are compact.

6.3 Compactness and Hausdorff

Theorem 6.3.1 ▶ Seperate a point and a compact set

Let X be Hausdorff. For any point $x \in X$ and any compact set $K \subseteq X\backslash\{x\}$ there exist disjoint open sets U and V with $x \in U$ and $K \subseteq V$.

Proof. □

Corollary 6.3.2 ▶ Compact subsets of Hausdorff spaces are closed

Compact subsets of Hausdorff spaces are closed.

Corollary 6.3.3 ▶ A map from compact to Hausdorff spaces is closed

If X is compact and Y is Hausdorff, then every map $f : X \to Y$ is closed.[1]
In particular,
- if f is injective, then it is an embedding;
- if f is surjective, then it is a quotient map;
- if f is bijective, then it is a homeomorphism.

6.4 Local Compactness

Definition 6.4.1

A space X is said to be ***locally compact at x*** if there is some compact subspace C of X that contains a neighborhood of x. If X is locally compact at each of its points, X is said simply to be ***locally compact***.

Theorem 6.4.2

Let X be a space. Then X is ***locally compact Hausdorff*** if and only if there exists a space Y satisfying the following conditions:

1. X is a subspace of Y.
2. The set $Y - X$ consists of a single point.
3. Y is a compact Hausdorff space.

If Y and Y' are two spaces satisfying these conditions, then there is a homeomorphism of Y with Y' that equals the identity map on X.

Proof. □

7 Metric Spaces

7.1 Convergence of Sequences

Definition 7.1.1 ▸ Convergence

Let (X, d) be a metric space and $x_1, \ldots, x_n, \ldots$ a sequence of points in X. Then the sequence is said to ***converge to*** $x \in X$ if given any $\varepsilon > 0$ there exists an integer n_0 such that for all $n \geq n_0$, $d(x, x_n) < \varepsilon$. This is denoted by $x_n \to x$.
The sequence $y_1, y_2, \ldots, y_n, \ldots$ of points in (X, d) is said to be ***convergent*** if there exists a point $y \in X$ such that $y_n \to y$.

Theorem 7.1.2

Let (X, d) be a metric space. $A \subseteq X$ is a subset, $p \in A$. Then p is a limit point of A, if and only if there exists a sequence $\{a_n\}$ composed with distinct points in A converging to p.

Proof. If p is a limit point of A. Then $B_d(p, 1) \cap (A \setminus \{p\}) \neq \varnothing$, we choose a point a_1 there. If we have points $a_1, a_2, \cdots, a_n$ such that $a_i \in B_d\left(p, \frac{1}{i}\right) \cap B_d(p, m_i)$, where

$$m_i := \min\left\{d\left(p, a_j\right)\right\}_{1 \leq j \leq i}$$

Since the sets $\left(B_d\left(p, \frac{1}{n+1}\right) \cap B_d(p, m_n)\right) \cap (A \setminus \{p\})$ is nonempty, we can take a point in it. We inductively give a seuquence $\{a_n\}$ composed with distinc points in A.

If $\{a_n\}$ is such sequence. Let U be any open set containg p. Since $\{B_d(x, \varepsilon) : x \in X, \varepsilon > 0\}$ is a basis for (X, d). We have from Proposition 1.2.4 that there exists a ball B such that $p \in B \subseteq U$. Then it is easy to verify that there exists $B_d(p, \varepsilon)$ such that $p \in B_d(p, \varepsilon) \subseteq B$. Since the sequence $\{a_n\}$ in A converges to p, there exists some (in fact, infinitely many) $a_n \in B_d(p, \varepsilon)$ that is not equal to p. i.e, $p \in B_d(p, \varepsilon) \cap (A \setminus \{p\}) \subseteq U \cap (A \setminus \{p\}) \neq \varnothing$. □

Proposition 7.1.3

Let (X, d) be a metric space. A subset A of X is closed in (X, d) if and only if every convergent sequence of points in A converges to a point in A. (In other words, A is closed in (X, d) if and only if $a_n \to x$, where $x \in X$ and $a_n \in A$ for all n, implies $x \in A$.)

7.2 Completeness

Definition 7.2.1 ▸ Cauchy sequence

A sequence $x_1, x_2, \dots, x_n, \dots$ of points in a metric space (X, d) is said to be a ***Cauchy sequence*** if given any real number $\varepsilon > 0$, there exists a positive integer n_0, such that for all integers $m \geq n_0$ and $n \geq n_0$, $d(x_m, x_n) < \varepsilon$.

Proposition 7.2.2

Let (X, d) be a metric space and $x_1, x_2, \dots, x_n, \dots$ a sequence of points in (X, d). If there exists a point $a \in X$, such that the sequence converges to a, that is, $x_n \to a$, then the sequence is a Cauchy sequence.

Definition 7.2.3

A metric space (X, d) is said to be ***complete*** if every Cauchy sequence in (X, d) converges to a point in (X, d).

Proposition 7.2.4

Let (X, d) be a metric space, Y a subset of X, and d_1 the metric induced on Y by d.

(i) If (X, d) is a complete metric space and Y is a closed subspace of (X, d), then (Y, d_1) is a complete metric space.

(ii) If (Y, d_1) is a complete metric space, then Y is a closed subspace of (X, d).

8 Countability and Speparation Axioms

8.1 The Countability Axioms

Definition 8.1.1 ▶ First-Countable Space

A space X is said to have a ***countable basis at x*** if there is a countable collection $\mathcal{B}$ of neighborhoods of x such that each neighborhood of x contains at least one of the elements of $\mathcal{B}$. A space that has a countable basis at each of its points is said to satisfy the ***first countability axiom***, or to be ***first-countable***.

Definition 8.1.2 ▶ Second-Countable Space

topological space X is called second-countable if it has a countable basis for its topology.

Definition 8.1.3 ▶ Lindelöf Space

A topological space X is called a Lindelöf space if every open cover of X has a countable subcover.

Theorem 8.1.4 ▶ Lindeöf Theorem

Every second-countable space is Lindeöf.

Proof. Let X be a second-coutable space, $\mathcal{B} = \{B_n\}_{n=1}^{\infty}$ is a countable basis of X. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be any open cover of X. Let

$$\mathcal{B}' := \{B_n \in \mathcal{B} : \exists U_\alpha \in \mathcal{U}, s.t B_n \subseteq U_\alpha\}$$

For each $x \in X$, $\exists U_\alpha \in \mathcal{U}$ such that $x \in U_\alpha$. Since $\mathcal{B}$ is a basis, there $\exists$ $B_n \in \mathcal{B}$ such that $x \in B_n \subseteq U_\alpha$. $B_n \in \mathcal{B}'$ by definition. Thus $\mathcal{B}'$ covers X. For each $B_{n_k} \in \mathcal{B}'$, we choose an open set U_k in $\mathcal{U}$, such that $B_{n_k} \subseteq U_k$. Then $\{U_k\}_{k=1}^{\infty}$ is a subcover of $\mathcal{U}$. □

9 Baire Space

Definition 9.0.1

A space X is said to be a ***Baire space*** if the following condition holds: Given any countable collection $\{A_n\}$ of closed sets of X each of which has empty interior in X, their union $\bigcup A_n$ also has empty interior in X.

Definition 9.0.2

A subset A of a space X was said to be of the ***first category*** in X if it was contained in the union of a countable collection of closed sets of X having empty interiors in X; otherwise, it was said to be of the ***second category*** in X.

Theorem 9.0.3

A space X is a ***Baire space*** if and only if every nonempty open set in X is of the ***second category***.

10

The Quotient Topology

Definition 10.0.1 ▸ Quotient Topology

Let $(X, \mathcal{T})$ and $(Y, \mathcal{T}_1)$ be topological spaces. Then $(Y, \mathcal{T}_1)$ is said to be a ***quotient space*** of $(X, \mathcal{T})$ if there exists a surjective mapping $f : (X, \mathcal{T}) \to (Y, \mathcal{T}_1)$ with the property (*).

$$\text{For each subset } U \text{ of } Y,\ U \in \mathcal{T}_1 \iff f^{-1}(U) \in \mathcal{T}. \qquad (*)$$

A surjective mapping f with the property(*) is said to be a ***quotient mapping***.

Note.
Trade-off: 陪域上的拓扑越精细，函数连续越困难；定义域越精细，函数连续越简单．我们为 f"施加压力"，使得它马上就变得不连续了，为的就是让连续函数 $g : X \to Z$ 可以安全且准确地下降为连续函数 $h : Y \to Z$，使得 $g = h \circ f$.

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